

Quiz 03. First-order ODEs in the wild
prepared by C. Cahig for ENS181 (AY 2026-2027, S1)

Instructions. In the following problems, the character d is reserved for derivatives and differentials so that $\frac{d\star}{d\omega}$ reads “the derivative of $\star$ with respect to ω ” and $d\alpha$ is “the differential of α ”. As is customary, e is the base of the natural logarithm, and $\exp(abc) = e^{abc}$.

Part I. Many an engineering system has parts with distinct thermal behaviours that to model the system’s thermal behaviour with just one temperature is to forgo fidelity. This can be addressed by treating the system as a *thermal network* where the temperatures of the parts are considered. A model for the time rate of change of temperature T_i of part i is

$$\dot{T}_i = \frac{\dot{Q}_i}{C_i} + \sum_{j \neq i} \frac{1}{R_{i,j}C_i} (T_j - T_i)$$

where¹

- T_j is the instantaneous temperature of part j ,
- C_i is the *thermal capacitance* of part i ,
- $R_{i,j}$ is the *thermal resistance* between parts i and j , and
- $\dot{Q}_i$ is the *internal rate of heat generation* of part i .

Consider now a CPU system having a *chip*, an *integrated heat spreader* (IHS²), and a *heatsink*. The chip has a *silicon die* that heats up noticeably during heavy computations (e.g., rendering). The IHS is a copper lid soldered onto the silicon die. The heatsink is a collection of aluminum sheets that absorb heat from the IHS and releases that heat to ambient air. Let T_c , T_s , T_h , and T_a denote the instantaneous chip, IHS, heatsink, and ambient-air temperatures (in °C). The chip, IHS, and heatsink have thermal capacitances C_c , C_s , and C_h equal to 1.5, 8.0, and 450 J/K, respectively³. Chip-IHS, IHS-heatsink, and heatsink-air connections have thermal resistances $R_{c,s}$, $R_{s,h}$, and $R_{h,a}$ equal to 0.1, 0.2, and 0.4 K/W, respectively. The CPU has been idling for a long time in 25 °C ambient air when it started a rendering task for which the chip draws 100 W from the power supply. Assume that the chip is practically thermally insulated from the IHS, from the heatsink, and from ambient air, and so is the HIS from ambient air. Further assume that the chip generates heat at the same rate it consumes electrical energy.

Scenario 1. Suppose the IHS temperature is known to be

$T_s(t) = 85 + 1.384e^{\lambda_1 t} - 20.520e^{\lambda_2 t} - 40.864e^{\lambda_3 t}$
where $\lambda_1 = -8.0224365$, $\lambda_2 = -0.5304573$, and $\lambda_3 = -0.0054395 \text{ s}^{-1}$. Derive the ODE for the chip temperature and write it in standard form $\dot{T}_c + p(t)T_c = q(t)$.

1. Determine $p(0)$.
2. Determine $p(-1/\lambda_1)$.
3. Determine $p(-1/\lambda_2)$.
4. Determine $p(-1/\lambda_3)$.
5. Determine $p(17)$.

¹ Analogous to electrical resistance, $R_{i,j}$ is a measure of how the physical connection between parts i and j hinders heat transfer. (This implies that $R_{i,j}$ is infinite if parts i and j are thermally insulated from each other.) Similarly, C_i is a measure of how much thermal energy part i can store per unit temperature change. $\dot{Q}_i = 0$ means part i “conveys” but does not generate heat. Ambient air can also be

6. Determine $q(0)$.
7. Determine $q(-1/\lambda_1)$.
8. Determine $q(-1/\lambda_2)$.
9. Determine $q(-1/\lambda_3)$.
10. Determine $q(17)$.
11. Determine $T_c(0)$.

Scenario 2. For the same IHS temperature as in Scenario 1, derive the ODE for the heatsink temperature, and write it in standard form $\dot{T}_h + r(t)T_h = v(t)$.

1. Determine $r(0)$.
2. Determine $r(-1/\lambda_1)$.
3. Determine $r(-1/\lambda_2)$.
4. Determine $r(-1/\lambda_3)$.
5. Determine $r(17)$.
6. Determine $v(0)$.
7. Determine $v(-1/\lambda_1)$.
8. Determine $v(-1/\lambda_2)$.
9. Determine $v(-1/\lambda_3)$.
10. Determine $v(17)$.
11. Determine $T_h(0)$.

Part II. In elementary electric circuit theory, capacitors and inductors are considered *energy storage elements*. For an ideal capacitor with capacitance C , its voltage v_C and current i_C is related given by

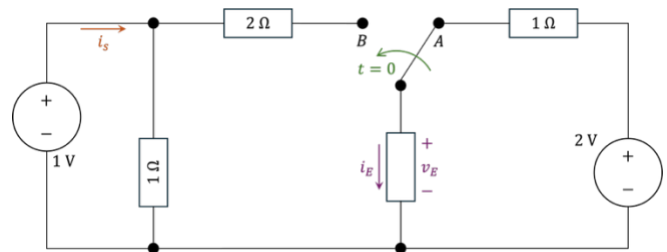
$$i_C(t) = C \frac{dv_C}{dt}$$

Similarly for an ideal inductor with inductance L , voltage v_L , and current i_L ,

$$v_L(t) = L \frac{di_L}{dt}$$

The theory imposes continuity on v_C and i_L . Graphically, there must not be a jump discontinuity when plotting the curves of v_C and i_L over time. Analytically, v_C and i_L must not abruptly change in (arbitrarily) infinitesimal time. Practically, the value of v_C (resp., i_L) just before a change in the circuit occurred, stays the same just after that change.

Consider the following circuit. The switch has been in position A for a long time (effectively fully-charging the storage element E) when it toggled to position B at $t = 0$.



treated as a “part” in the thermal network, since it allows heat transfer via convection.

² This is not a Jesuit plant. Or is it?

³ To convert a temperature measured in °C to K, just add 273.15 to the numerical value. Knowing this and that thermal capacitance is energy per change in temperature, a thermal capacitance measured in J/K is numerically equivalent to that in J/°C.

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Scenario 3. Suppose the storage element is a 1-mF capacitor and whose full-charge voltage is 2 V. Derive the first-order ODE involving $\dot{v}_E$, and solve for the capacitor voltage $v_E(t)$. With $v_E(t)$, you can readily solve for the capacitor current $i_E(t)$. Knowing $v_E(t)$ and $i_E(t)$, you can use Kirchhoff's and/or Ohm's laws to solve for the current $i_s(t)$.

1. Is the ODE linear in v_E ?
2. Determine $v_E(0)$.
3. Determine $v_E(\tau)$, where $\tau = 2 \times 10^{-3}$.
4. Determine $v_E(3\tau)$.
5. Determine $v_E(5\tau)$.
6. Determine $v_E(100\tau)$.
7. Determine $i_E(0)$.
8. Determine $i_E(\tau)$.
9. Determine $i_E(3\tau)$.
10. Determine $i_E(5\tau)$.
11. Determine $i_E(100\tau)$.
12. Determine $i_s(0)$.
13. Determine $i_s(\tau)$.
14. Determine $i_s(3\tau)$.
15. Determine $i_s(5\tau)$.
16. Determine $i_s(100\tau)$.

Scenario 4. Suppose instead that the storage element is a 1-mH inductor and whose full-charge current is 2 A. Derive the first-order ODE involving $\frac{di_E}{dt}$, and solve for the inductor current $i_E(t)$. With $i_E(t)$, you can readily solve for the inductor voltage $v_E(t)$. Knowing $v_E(t)$ and $i_E(t)$, you can use Kirchhoff's and/or Ohm's laws to solve for the current $i_s(t)$.

1. Is the ODE linear in i_E ?
2. Determine $i_E(0)$.
3. Determine $i_E(\tau)$, where $\tau = 0.5 \times 10^{-3}$.
4. Determine $i_E(3\tau)$.
5. Determine $i_E(5\tau)$.
6. Determine $i_E(100\tau)$.
7. Determine $v_E(0)$.
8. Determine $v_E(\tau)$.
9. Determine $v_E(3\tau)$.
10. Determine $v_E(5\tau)$.
11. Determine $v_E(100\tau)$.
12. Determine $i_s(0)$.
13. Determine $i_s(\tau)$.
14. Determine $i_s(3\tau)$.
15. Determine $i_s(5\tau)$.
16. Determine $i_s(100\tau)$.

Marking. Each numbered item above is worth two (2) points. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.